Referee Report: "Consumer Information and the Limits to Competition"

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American Economic Review, 2022, 112(2), 534–577

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April 14, 2026

1. Summary

1.1. Motivation

In many markets, firms supply differentiated products and compete in prices, while consumers possess incomplete information about how well each product matches their preferences. A central question for platforms (e.g., Amazon, Expedia, insurance comparison websites) and regulators is how to design the information environment that consumers face.

The key tension is immediate: more transparency improves product-consumer matching, but it also softens price competition by giving firms greater market power. Less transparency forces price competition — products are perceived as near-perfect substitutes — but consumers may end up purchasing a poorly matched product.

Armstrong and Zhou formalize this trade-off. They ask: within the class of all feasible signal structures, which one maximizes industry profit, and which one maximizes consumer surplus? Their answer reveals that the interests of firms and consumers are clearly opposed in how information should be structured, yet both sides agree that information should be symmetric across firms.

1.2. Model Essentials

Two symmetric firms ($i = 1, 2$) supply differentiated products at zero cost. A risk-neutral consumer values product i at $v_i \geq 0$; only her relative preference $x \equiv v_1 - v_2$ matters. Let F denote the prior CDF of x, symmetric on $[-1, 1]$, with density f. Two key parameters summarize the prior:

- $\mu \equiv E[v_i]$: expected value of a random product.
- $\delta \equiv E_F[\max\{x, 0\}] = \int_{-1}^{0} F(x)\,dx$: expected match-quality gain from buying the preferred product over a random one.

Before buying, the consumer observes a private signal s from a signal structure $\{\sigma(s|x), S\}$ chosen by a designer —this is common knowledge. She updates to a posterior; since she is risk-neutral, only the posterior mean ($E[x \mid s]$) matters. Let G denote the CDF of this posterior mean — the posterior distribution. Firms observe G but not s, and set prices simultaneously (Bertrand–Nash). The consumer buys from firm 1 if $E[x \mid s] > p_1 - p_2$.

The only constraint on G is Bayesian consistency: G must be a mean-preserving contraction (MPC) of F[1]. Any such G can be generated by some signal structure. So instead of optimizing over signals, the authors optimize directly over posteriors — a key simplification.

Lemma 1 shows that a posterior G can sustain symmetric equilibrium price p iff G lies between two bounds derived from each firm's no-deviation condition[2]:

$$L_p(x) \;\leq\; G(x) \;\leq\; U_p(x)$$

Intuitively, for the upper bound, higher p flattens both bounds, requiring fewer near-indifferent consumers to avoid firms being able to deviate and attract all consumers. **Here I need only to explain the intuitiion behinf the upper bound, for increased... Simply the distribution can not have certain characteristcs that make deviations always profitable.**

1.3. Key Insight: Symmetric Signals Dominate

Before solving the optimal design, the paper establishes a powerful restriction: for any asymmetric signal (one that favors a firm[3]), there exists a symmetric signal that yields higher profit for *each* firm and higher consumer surplus. Biased recommendations trigger aggressive undercutting by the disfavored firm, and this ends up hurting everyone. This means we can restrict attention to symmetric posteriors and symmetric prices throughout —which is the second big simplification in the paper.

[1], i.e.,

$$\int_{-1}^{x} G(\tilde{x})\,d\tilde{x} \;\leq\; \int_{-1}^{x} F(\tilde{x})\,d\tilde{x} \qquad \forall\, x \in [-1, 1].$$

[2]Note this is only in pure strategies.

[3]We can think of this as a biased signal that tells consumers that one good is much better than the other.

1.4. Main Results

Firm-optimal signal. The firm-optimal posterior sits on the *lower* bound of the feasibility corridor, yielding a U-shaped density: consumers are pushed toward the extremes, few are near-indifferent, and neither firm can profitably undercut (this is why it bounds). The resulting price is

$$p^* = \frac{2\delta}{1 - \log 2} \approx 6.5\,\delta,$$

where $\delta \equiv E_F[\max\{x, 0\}]$ measures underlying product differentiation. **Comment, I don't know if we are interested in shoing this exact result, unless the inntuition of the result is conveyed... it is worthless without intutiion, which I provide below.**Firms want highest prices, so firms set higher differentiation st to no deviation–¿ make the bounds bind. 2 results are key: first, there is **no mismatch**; the consumer buys her preferred product, and total welfare is maximized. Second, note that for firms it is not optimal to provide full information: it precisely needs to exaggerate preferences - while not changing relative valuation for good (one vs the other)- but extending perceived differentiation so firms can charge higher prices.

Consumer-optimal signal. The consumer-optimal posterior is more tricky: consumers want higher prices - so no differnetiation- but also want their match product. They show tht prices matter more and that the optimal distribution rides the *upper* bound: mass is concentrated near $x = 0$, creating many swing consumers who make firms compete fiercely. The price drops to $p^{**} \approx 0.05\,p_F$. The cost is **mismatch** — some consumers buy the wrong product — but the price effect dominates. Importantly, for consumders the best solution is not full no information disclosure: some information improves matches and does not relax price competition.

I think this is not important at all, main results are above... The limits to competition. Any profit $p \in [0, p^*]$ is feasible. Competition makes it *impossible* to have both low profit and low consumer surplus simultaneously — unlike the monopoly setting of Roesler and Szentes (2017). The feasible set has a concave frontier, and full disclosure is optimal only for unweighted total welfare.

1.5. Application

I think this is completely redundant: Two extreme cases make the tension that the papers explores concrete. Under **full disclosure** ($G = F$), the consumer always buys her preferred product (max welfare), but firms enjoy market power and charge high prices. Under **no disclosure** (G degenerate at zero), products look identical, Bertrand competition drives prices to zero, but the consumer buys randomly and may get the wrong product. Surprising neither of the two extremes is optimal for firms and consumers: consumers benefit from some degree of information, because prices dont change much and

matches are increased (vs no information); firms benefit from incorporating not perfect information: basically enlarging perceived differentiation...

The paper's contribution is to characterize the *entire* spectrum between these extremes: which posteriors G maximize firm profit, which maximize consumer surplus, and what combinations of profit and surplus are feasible.

1.6. Extensions

Maybe describe a bit what are the two extensions they contemplate in their paper. But I think providing the main message is enough. Why they both pull to the extremes and wyy it binds (equilibrium condition; intuitively some distributions dont sustain deviations.) what happens in each equilibrium that is optimal for firms and for consumders- This is the key result of the paper.

2. Critique

Some comments on the limitations of the paper

2.1. Scalar Heterogeneity and the Duopoly Restriction

The paper reduces the consumer's two-dimensional preference (v_1, v_2) to a scalar relative valuation $x = v_1 - v_2$. This is natural in the duopoly case, but it means that the *level* of consumer willingness-to-pay (captured by $V = v_1 + v_2$) is effectively assumed away. In practice, both the level and the relative preference matter: a consumer who values both products highly behaves differently from one who values both poorly, even if their relative preference x is identical. With more than two firms, preferences are inherently multidimensional, and the paper's approach does not extend. The authors study three simple policies ("top product," "top two," and full disclosure) for $n > 2$, but the full characterization of optimal signal structures remains open.–¿ Com: I think here we need to take a subset, can not give all the critiques... Choose the top.

2.2. Pure-Strategy Restriction

The focus on signal structures that induce pure-strategy pricing equilibria is motivated by tractability. Proposition 4 shows that allowing mixed strategies improves consumer surplus by at most 1.6%, which is reassuring. However, the firm-optimal signal under mixed strategies remains uncharacterized. More importantly, for some prior distributions and signal structures, pure-strategy equilibria may not exist, and the paper's framework provides no guidance for these cases. Understanding whether the qualitative insights — symmetric dominance, the shape of the feasible frontier — survive under mixed strategies would strengthen the paper's conclusions. –¿is this such a problem? they justify this, so

I think this is bullshit. Pure strategies is closed to a behavioral assumtpion, why should they randomize?

2.3. Exogeneity of the Signal Structure

Comm: i think this is the strongest critique, but feel free to disagree. The entire analysis assumes the signal structure is chosen exogenously by a designer and committed to before firms set prices. This "centralized design" assumption is strong. In many real markets, information is generated through a combination of firm advertising, consumer search, platform algorithms, and third-party reviews.

- If firms can influence the signal structure (e.g., through advertising), the optimal centralized policy may not be implementable. Comm why?
- If consumers can choose how much information to acquire (endogenous search), the designer's ability to control the posterior is limited. True!
- The timing assumption is crucial. If firms could respond to the signal structure by adjusting product characteristics or quality, the welfare conclusions could reverse. Comm: for example: if little infomration, increasing quality? but would consumders perceive it? I like more the idea of the advertising outside the platform: if little within info, then advertise more outside. This could also be an extension... s

2.4. Symmetry of the Prior

Comm:This is also nice, tight to advertising campaigns. The paper assumes firms are ex ante symmetric: the prior over (v_1, v_2) is symmetric between the two products. Many real-world markets feature vertical differentiation (one product is known to be better on average) or brand effects. The authors suggest (Section IV) that extending to asymmetric priors is interesting, but they do not explore whether the optimality of symmetric signals (Lemma 2) survives. With an asymmetric prior, biased signals that further favor the better firm might no longer be dominated, and the structure of the feasible set could change qualitatively.

2.5. Welfare Metric and Dynamic Considerations

Comm: This is also nice: fidelity, consumer satisfaction etc... but here consumers are still satisfied due to prices, no?

The consumer-optimal signal structure deliberately introduces mismatch: some consumers buy their less-preferred product. The paper treats this as a pure static efficiency loss, measured by $\delta - \int_{-1}^{0} G(x)\,dx$. However, in practice, product mismatch may generate dynamic costs — consumer regret, returns, negative reviews, reduced repeat purchases — that are not captured in the one-shot model. If these costs are substantial, the consumer-optimal policy may be less attractive than the static analysis suggests, and a

regulator might optimally choose a signal structure that lies between the firm-optimal and consumer-optimal extremes.

3. Extension: Platform as Information Designer

Here we could provide more intuitively, without the math, the intuition behind two models. Why and what would change and then how would the predictions look like in this new setting.

3.1. Motivation

Armstrong and Zhou note that "it would be valuable to embed this analysis within a framework in which the 'information designer' is modeled explicitly as an economic agent" (Section IV, p. 562). In their paper, signal structures are chosen to maximize firm profit or consumer surplus, but no agent actually performs this optimization. A natural candidate is a **platform** — Amazon, Expedia, an insurance comparison site — that controls how much product information consumers see and monetizes through fees charged to firms or consumers.

The question is simple: *what signal structure would a profit-maximizing platform choose, and how does its fee structure shape that choice?* Using the paper's existing toolkit — the bounds L_p, U_p, the MPC constraint, and the feasible set from Proposition 3 — we can trace through the answer under three fee regimes. The results suggest that the platform's business model is not a distributional detail: it determines the entire information environment.

3.2. Setup

We add a monopoly platform to the Armstrong–Zhou duopoly. The timing is:

1. The platform commits to a signal structure (posterior G) and a fee schedule.
2. Firms observe G and the fees, decide whether to participate, and set prices simultaneously.
3. The consumer observes her private signal s, pays the price (plus any per-unit fee), and buys one product.

Everything else is unchanged: two symmetric firms, zero marginal cost, prior F on $[-1, 1]$, consumer always buys. We consider three fee instruments:

- **Per-transaction fee** τ: the platform collects τ on every sale.
- **Lump-sum listing fee** ϕ: each firm pays ϕ upfront to be listed.
- **Mixed fees** (τ, ϕ): the platform uses both instruments simultaneously.

For each case, the platform chooses (G, fees) to maximize its revenue, subject to: (i) G is an MPC of F, (ii) G lies in $[L_p, U_p]$ for some symmetric equilibrium price p,

(iii) each firm's participation constraint holds (net profit ≥ 0), and (iv) the consumer's participation constraint holds ($CS \geq 0$).

3.3. Pass-Through of Per-Unit Fees

A key observation simplifies the per-unit case. With fee τ, each firm's effective cost is τ. Define the markup $\tilde{p}_i \equiv p_i - \tau$. Since the consumer's choice depends only on the price *difference* $p_1 - p_2 = \tilde{p}_1 - \tilde{p}_2$, and each firm maximizes $\tilde{p}_i \cdot q_i(\tilde{p}_1 - \tilde{p}_2; G)$, the pricing game in markups is identical to the original zero-cost game. The no-deviation bounds L_p, U_p depend only on the price difference and the symmetric markup level, so they are unchanged. The equilibrium markup is therefore $\tilde{p} = p(G)$, the consumer pays $p(G) + \tau$, and each firm's gross profit remains $p(G)/2$ regardless of τ. The per-unit fee is **fully passed through** to consumers — a consequence of inelastic total demand and price-difference-based allocation.

3.4. Case 1: Per-Unit Fees Only

The platform earns τ per sale ($= \tau$ total, since one unit is always sold). It wants τ as large as possible. The binding constraint is consumer participation. Under posterior G and fee τ, the consumer's surplus is:

$$CS \;=\; \underbrace{\mu + \int_{-1}^{0} G(x)\,dx}_{\text{expected gross utility}} \;-\; \underbrace{\big(p(G) + \tau\big)}_{\text{total price paid}}.$$

Setting $CS = 0$ (extracting all consumer surplus) gives the platform's optimal fee:

$$\tau^* \;=\; \mu + \int_{-1}^{0} G(x)\,dx \;-\; p(G).$$

The right-hand side is exactly $CS(G)$ from the original Armstrong–Zhou game. So the platform's problem reduces to: *choose G to maximize consumer surplus in the baseline game.* By Proposition 2 of the paper, the solution is the consumer-optimal signal $G^{**} = \min\{F, U_{p^{**}}\}$ with $p^{**} \approx 0.05\, p_F$.

Outcome: the consumer-optimal signal is implemented, but consumers get *zero* surplus — τ extracts it all. Firms earn small positive markups $p^{**}/2$. Mismatch occurs, so total welfare falls below $\mu + \delta$.

3.5. Case 2: Lump-Sum Fees Only

The platform charges ϕ per firm upfront. Since ϕ does not enter marginal cost, it does not affect pricing or the consumer's purchase decision. Each firm participates if and only

if its gross profit covers the fee: $p(G)/2 \geq \phi$. The platform sets $\phi^* = p(G)/2$ and earns $2\phi^* = p(G)$.

The platform's problem reduces to: *choose G to maximize industry profit $p(G)$*. By Proposition 1 of the paper, the solution is the firm-optimal signal $G^* = L_{p^*}$ with $p^* = 2\delta/(1 - \log 2) \approx 6.5\,\delta$.

Outcome: the firm-optimal signal is implemented. No mismatch — total welfare is maximized at $\mu + \delta$. Consumers retain surplus $CS = \mu + \delta - p^* > 0$. (For the uniform prior: $\mu = 2$, $\delta = 1/4$, $p^* \approx 1.63$, so $CS \approx 0.62$.)

3.6. Case 3: Mixed Fees

When the platform can charge both τ and ϕ, both participation constraints bind at the optimum: τ extracts all consumer surplus and ϕ extracts all firm profit. The platform's total revenue is:

$$\begin{aligned}
\Pi_{\text{platform}} &= \tau + 2\phi \\
&= \underbrace{\left(\mu + \int_{-1}^{0} G(x)\,dx - p(G)\right)}_{\tau^* = CS(G)} + \underbrace{p(G)}_{2\phi^* = p(G)} \\
&= \mu + \int_{-1}^{0} G(x)\,dx.
\end{aligned}$$

The equilibrium price $p(G)$ cancels. The platform's revenue depends only on the match efficiency $\int_{-1}^{0} G(x)\,dx$, which is maximized at δ when G induces no mismatch. Any signal that ensures the consumer buys her preferred product achieves this — including full disclosure ($G = F$) or the firm-optimal signal ($G = L_{p^*}$). The platform earns $\mu + \delta$, which equals maximum total welfare. Both consumers and firms are left with zero surplus.

Outcome: the platform becomes the residual claimant on total welfare and therefore maximizes efficiency. The signal choice is pinned down only up to the no-mismatch condition — the specific posterior does not matter, only that the consumer always buys her preferred product. (For the uniform prior: platform revenue = 2.25, $CS = 0$, firm profit = 0.)

3.7. Comparison and the Fee-Structure Paradox

	Per-unit τ	**Lump-sum** ϕ	**Mixed** (τ, ϕ)
Platform maximizes	CS	Industry profit	Total welfare
Signal chosen	G^{**} (cons-opt)	L_{p^*} (firm-opt)	Any no-mismatch G
Mismatch?	Yes	No	No
Total welfare	$< \mu + \delta$	$= \mu + \delta$	$= \mu + \delta$
Consumer surplus	0	$\mu + \delta - p^* > 0$	0
Firm profit (net)	$p^{**}/2 > 0$	0	0
Platform revenue	$\mathrm{CS}(G^{**})$	p^*	$\mu + \delta$

The most striking result is the **fee-structure paradox**: consumers are best off under lump-sum fees, which induce the *highest* equilibrium prices. The mechanism is that the platform captures surplus from whichever side it taxes. Under per-unit fees, it taxes consumers, so it designs information to maximize consumer surplus — then takes it all. Under lump-sum fees, it taxes firms, so it designs information to maximize firm profits — and consumers keep the untaxed residual $\mu + \delta - p^*$. The "consumer-friendly" fee structure (commissions) leads to worse consumer outcomes than the "firm-friendly" one (listing fees).

3.8. Takeaways

The business model shapes information design. The platform's fee structure is not a mere distributional detail — it determines which signal the platform wants to implement. Per-unit fees align the platform's incentive with consumer surplus (which it then extracts); lump-sum fees align it with firm profit. Mixed fees align it with total welfare. This echoes a theme from Rochet and Tirole (2003): in two-sided markets, the *structure* of fees, not just their level, has allocative consequences.

Regulatory implications. A regulator who cannot directly control information design might regulate the fee structure instead. Mandating lump-sum fees indirectly implements the welfare-maximizing signal *and* leaves consumers with positive surplus. Allowing unrestricted mixed fees achieves efficiency but leaves consumers with nothing. Mandating per-unit fees is the worst option: consumers get zero surplus *and* total welfare is reduced by mismatch.

Limitations. This sketch assumes a monopoly platform with full commitment. With competing platforms, fees would be driven down and information design incentives would weaken. Without commitment (the platform adjusts G after observing prices), the timing

game changes fundamentally. Finally, if the platform chooses its own fee structure, it picks mixed fees — achieving efficiency but extracting all surplus.

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A. Mathematical Details for the Platform Extension

This appendix collects the step-by-step calculations behind the results in Section 3. All notation follows Armstrong and Zhou (2022): F is the prior CDF of $x = v_1 - v_2$ on $[-1, 1]$; G is the posterior CDF; $p(G)$ is the symmetric equilibrium price under posterior G with zero cost; $\mu = E[v_i]$; $\delta = \int_{-1}^{0} F(x)\,dx$; $CS(G) = \mu + \int_{-1}^{0} G(x)\,dx - p(G)$.

A.1. Pass-Through of Per-Unit Fees

With per-unit fee $\tau \geq 0$, firm i's profit is $(p_i - \tau)\, q_i(p_1 - p_2; G)$. Define the markup $m_i \equiv p_i - \tau$. Then:

- Firm i's profit $= m_i \cdot q_i(m_1 - m_2; G)$, since $p_1 - p_2 = m_1 - m_2$.
- The demand function q_i depends on G and the price difference only.
- Firm i's first-order condition in m_i is identical to the zero-cost first-order condition in p_i.

For the no-deviation bounds, recall that in the original game, firm 2's no-deviation condition requires $p_2'\, G(p_1 - p_2') \leq p \cdot G(0) = p/2$ for all p_2'. With cost τ, this becomes $(p_2' - \tau)\, G(p_1 - p_2') \leq (p - \tau)/2$ for all p_2'. Substituting $m_2' = p_2' - \tau$ and $m = p - \tau$:

$$m_2'\, G(m + \tau - m_2' - \tau) = m_2'\, G(m - m_2') \leq m/2.$$

This is the same bound as in the zero-cost game with symmetric markup m. Therefore, the equilibrium bounds L_p, U_p apply to the markup $m = p(G)$, the equilibrium retail price is $p(G) + \tau$, and each firm's profit is $p(G)/2$.

A.2. Platform's Problem: Per-Unit Fees

The platform chooses G and τ to maximize τ, subject to:

$$G \text{ is an MPC of } F, \tag{1}$$

$$L_{p(G)}(x) \leq G(x) \leq U_{p(G)}(x) \quad \forall\, x, \tag{2}$$

$$\underbrace{\mu + \int_{-1}^{0} G(x)\,dx - p(G) - \tau}_{\text{consumer surplus}} \geq 0, \tag{3}$$

$$p(G)/2 \geq 0. \tag{4}$$

Constraint (4) is trivially satisfied. Constraint (3) binds at the optimum (the platform extracts as much as possible):

$$\tau^* = \mu + \int_{-1}^{0} G(x)\,dx - p(G) = CS(G).$$

The platform's problem becomes $\max_G CS(G)$ subject to (1)–(2). This is exactly the consumer's problem in Armstrong–Zhou. By their Proposition 2, the solution is:

$$G^{**}(x) = \min\left\{F(x),\ U_{p^{**}}(x)\right\} \quad \text{for } x < 0,$$

with p^{**} determined by $\gamma \approx 0.05$ as in the paper.

Numerical check (uniform prior): $\mu = 2$, $\delta = 1/4$, $p_F = 1$, $p^{**} \approx 0.05$. The consumer-optimal CS in the baseline game is approximately $\mu - p^{**} +$(match value under G^{**}) $\approx$ 2.20. The platform sets $\tau^* \approx 2.20$ and earns this amount. Firms earn $p^{**}/2 \approx 0.025$ each.

A.3. Platform's Problem: Lump-Sum Fees

The platform chooses G and ϕ to maximize 2ϕ, subject to:

$$G \text{ is an MPC of } F, \tag{5}$$

$$L_{p(G)}(x) \le G(x) \le U_{p(G)}(x) \quad \forall\, x, \tag{6}$$

$$\mu + \int_{-1}^{0} G(x)\,dx - p(G) \ge 0, \tag{7}$$

$$p(G)/2 - \phi \ge 0. \tag{8}$$

Constraint (8) binds: $\phi^* = p(G)/2$, so revenue $= p(G)$. The problem becomes $\max_G p(G)$ subject to (5)–(6). This is the firm's problem in Armstrong–Zhou. By their Proposition 1:

$$G^* = L_{p^*}, \qquad p^* = \frac{2\delta}{1 - \log 2}.$$

Since $G^* = L_{p^*}$ induces no mismatch, $\int_{-1}^{0} G^*(x)\,dx = \delta$, and consumer surplus is:

$$CS = \mu + \delta - p^* = \mu + \delta - \frac{2\delta}{1 - \log 2} = \mu - \frac{(1 + \log 2)\,\delta}{1 - \log 2}.$$

This is positive whenever $\mu > \frac{(1+\log 2)}{(1-\log 2)}\,\delta \approx 5.5\,\delta$. Under the paper's maintained assumption $V > 2$ (which implies $\mu \ge 2$), and since $\delta \le 1/4$ for the uniform prior, this holds easily: $5.5 \times 0.25 = 1.375 < 2$.

Numerical check (uniform prior): $p^* \approx 1.63$, $\phi^* \approx 0.815$, platform revenue ≈ 1.63, $CS = 2 + 0.25 - 1.63 = 0.62$.

A.4. Platform's Problem: Mixed Fees

The platform chooses G, τ, and ϕ to maximize $\tau + 2\phi$, subject to (1)–(2) and:

$$\mu + \int_{-1}^{0} G(x)\,dx - p(G) - \tau \geq 0, \tag{9}$$

$$p(G)/2 - \phi \geq 0. \tag{10}$$

Both constraints bind at the optimum: $\tau^* = \mu + \int_{-1}^{0} G(x)\,dx - p(G)$ and $\phi^* = p(G)/2$. Therefore:

$$\begin{aligned}\Pi_{\text{platform}} &= \tau^* + 2\phi^* \\ &= \Big(\mu + \int_{-1}^{0} G(x)\,dx - p(G)\Big) + p(G) \\ &= \mu + \int_{-1}^{0} G(x)\,dx.\end{aligned}$$

The price $p(G)$ drops out. The platform's revenue equals the consumer's expected gross utility, which is maximized when $\int_{-1}^{0} G(x)\,dx = \delta$ (no mismatch). The maximum revenue is $\mu + \delta$.

Any posterior with no mismatch achieves this: $G = F$ (full disclosure), $G = L_{p^*}$ (firm-optimal), or any G such that the consumer buys her preferred product with probability one. The platform is indifferent among all of them.

Numerical check (uniform prior): $\Pi_{\text{platform}} = 2 + 0.25 = 2.25$. Under full disclosure ($p_F = 1$): $\tau^* = 2 + 0.25 - 1 = 1.25$, $\phi^* = 0.5$, total $= 1.25 + 1.0 = 2.25$. ✓